

Baoren Liu

✉ liubaoren2006@gmail.com ☎ +1 781-457-8199 🌐 baorenliu.com in baoren-liu-a27141344 🤖 RoboticReaper

Education

University of Illinois Urbana-Champaign, IL
B.S. in Mathematics and Computer Science

Expected May 2028
GPA 3.92/4.00

Experience

AI Research Intern

June 2024 - Present

Mass General Brigham [medRxiv.org preprint](https://medrxiv.org/preprint)
Boston, MA

- Processed **115+ million** proprietary health records in a **HIPAA-compliant** environment, deploying a **RAG** pipeline to identify dementia patients. *Tools: Python, Azure, ChromaDB, PostgreSQL (pgvector), LangChain, PyTorch, Pandas, NumPy*
- Achieved a **0.933 F1-score** by optimizing RAG retrieval (MMR, prompt engineering) and evaluating advanced open-weight LLMs and biomedical embedding models using Ollama and HuggingFace
- Benchmarked RAG pipelines against traditional ML models (**XGBoost, SVM, RF, LR**), validating pipeline performance via cross-validation, bootstrapping, and plotting visualizations
- Co-authored** a research paper evaluating these methodologies, currently under peer-review in *Journal of Biomedical Informatics*
- Automated deployment of the [research website](#) through **GitLab CI/CD** and **SSH-managed Linux** runners, eliminating manual deployment and enabling continuous updates. *Tools: Plesk webhook, AI assistants (Claude, Gemini, Antigravity)*

Founder & Project Leader & Technical Lead

August 2020 - May 2024

Lexington Youth STEAM Team youthsteaminitiative.org
Lexington, MA

- Partnered with local municipal committees to deploy a virtual event platform for Patriots' Day during COVID-19, processing **200+** community media submissions. *Tools: WordPress, HTML/CSS/JS*
- Conducted a historical grant trend analysis for the Community Endowment of Lexington, identifying key community needs to facilitate a record **\$57,850** funding cycle. *Tools: Pandas, NumPy*
- Directed volunteer recruitment, onboarding, and continuous maintenance of the organization's website across **400+** hours of dedicated community service. *Tools: GitHub Pages, HubSpot*

Projects

RideList ridelist.app

January 2026 - Present

- Built and launched a bilingual (English/Chinese) campus ridesharing web app using Next.js and Mantine UI, supporting **60+ trip postings, 75+ in-app messages, and 7 completed ride bookings**
- Designed a normalized PostgreSQL schema and optimized database access patterns using indexes, joins, and CTEs to efficiently support ride discovery, messaging, and trip management workloads
- Implemented a certificate-based security encryption using a custom **Certificate Authority** and **mTLS** to mutually authenticate communications between Vercel API servers and self-hosted database. *Other tools: Cloudflare API, DDNS*
- Developed a privacy-preserving geospatial trip-matching system using **PostGIS** and the **Google Maps/Places API** to enable fuzzy location sharing while maintaining accurate ride discovery
- Built real-time messaging, trip state synchronization, and automated remote PostgreSQL backups using Firebase Cloud Messaging, pg_dump, cron, and rsync

Schedule Web App lhsschedule.netlify.app

July 2020 - December 2024

- Developed a successful high school scheduling web app, seamlessly handling complex real-time schedule adjustments and features like multi-device syncing, offline-loading, PDF-scanning, and friends. *Tools: ReactJS, Material UI, Firebase*
- Supported **1,400+** registered and **900** daily active users consistently during peak back-to-school traffic by scaling on **Netlify**

Awards

- 1st Place (Beginner) & 6th/131 Overall (Advanced)** at UIUC SIGPwny Fall 2025 Capture The Flag Cybersecurity Competition 2025.fallctf.com ([team null](#))
September 21, 2025

Skills

Languages: Python, Java, C++, SQL, HTML/CSS, JavaScript, TypeScript, Kotlin, Bash

Frameworks & Tools: PyTorch, LangChain, React, Next.js, Docker, PostgreSQL, pgvector, PostGIS, ChromaDB, Git, GitLab CI/CD, Azure, Firebase, Linux, Tailscale, Android Studio